

Elicitation versus Teaching: Why Verifiable-Reward Data Compounds Most Where Latent Skill Is Absent

Perpetual Labs Research

Perpetual Labs — Applied AI Research

Research Note PL-RN-2026-i07 · correspondence: research@perpetual-labs.ai

Abstract

Reinforcement Learning with Verifiable Rewards (RLVR) is the dominant post-training recipe behind frontier reasoning models, yet a 2025 body of evidence shows its gains are frequently elicitation of latent capability rather than the teaching of new capability. RLVR improves Qwen2.5-Math-7B on MATH-500 by +21.4 points with random rewards and +24.1 points with deliberately incorrect labels — within ~8 points of the +29.1 from ground-truth rewards [1] — while the identical recipe yields minimal or negative gains on Llama3 and OLMo2 [1]. A parallel debate finds RLVR narrows the pass@k reasoning boundary at large k on Qwen families [2], even as prolonged RL (ProRL) and CoT-Pass@K analyses demonstrate genuine boundary expansion on tasks the base model cannot reach at any sampling budget [3][4]. We synthesize these results into a data-strategy thesis: where capability is already latent (math/code on Qwen-class bases), verifiable-reward data largely reveals existing skill and its marginal value is low; where capability is absent — specialized, under-documented, proprietary verticals — verifiable data genuinely teaches, as Med-RLVR's +8-point out-of-distribution gain over SFT illustrates [5]. We formalize an elicitation-vs-teaching diagnostic, quantify it across model families, and argue the highest-value verifiable-reward environments target genuine knowledge frontiers, not saturated public benchmarks.

1. Introduction

Reinforcement Learning with Verifiable Rewards (RLVR) — training a policy against an automatic checker that grades final answers as pass/fail — has become the core recipe behind frontier reasoning systems, from DeepSeek-R1 [6] to open replications [7] and, by late 2025, to scaled-RL frontier models: DeepSeek-V3.2-Special reaches gold-medal performance at IMO 2025 (35/42) and IOI 2025 (492/600) purely by scaling post-training RL compute, surpassing GPT-5 and reaching reasoning parity with Gemini-3.0-Pro [14]. The prevailing narrative treats RLVR as a capability-creation engine: point a verifiable reward at a base model and reasoning emerges. If true, this would imply that the scarce input is the reward checker (the algorithm), and that any domain with a checker can be uplifted regardless of what the base model already knows.

A cluster of 2025 results complicates this narrative. Shao et al. show that RLVR with random, format-only, or even deliberately incorrect rewards still lifts Qwen2.5-Math-7B on MATH-500 by double-digit margins — but that the same spurious signals fail, or actively hurt, on Llama3 and OLMo2 [1]. Yue et al. show that RLVR-trained models beat their base at pass@1 yet are overtaken by the base at large k, suggesting RLVR sharpens sampling rather than expanding the reasoning frontier [2]. Together these imply that much of what RLVR 'achieves' on popular math/code benchmarks is elicitation of capability the base model already latently holds.

This distinction — eliciting versus teaching — is not a methodological footnote. It is the hinge of a data-acquisition strategy. If verifiable-reward data mostly reveals latent skill, then building expensive verifiable environments in domains the base model has already absorbed (public math, public code) buys little durable

capability. The data that genuinely teaches — and therefore compounds into a defensible asset — is concentrated where the base model has no latent skill to elicit: specialized, proprietary, under-documented vertical workflows. This paper makes that inference explicit, grounds it in real benchmark numbers, and converts it into a vertical-selection diagnostic.

Our contribution is threefold. First, we assemble the elicitation evidence and the counter-evidence (ProRL, CoT-Pass@K) into one quantified picture rather than treating them as opposing camps. Second, we formalize an elicitation-vs-teaching score from measurable quantities (cross-family transfer of spurious rewards, pass@k crossover, base-model zero-coverage rate). Third, we draw the data-strategy consequence: target verifiable verticals where the teaching component dominates.

2. Background and Related Work

RLVR descends from a line of work establishing that verifiable, process-level supervision beats outcome labels for hard reasoning. Lightman et al. showed a process-supervised reward model solves 78% of a representative subset of the MATH test set and significantly outperforms outcome supervision and majority voting, releasing PRM800K (800K step-level human labels) [8]. The verifiable-reward framing was named and scoped to 'tasks with checkable answers' in 2024 [9], and DeepSeek-R1-Zero demonstrated that reasoning behaviors — self-reflection, verification, strategy adaptation — can emerge from pure RL with no supervised fine-tuning [6].

Open systems rapidly matched the closed frontier on these verifiable tasks. DAPO reaches 50 points on AIME 2024 with a Qwen2.5-32B base, exceeding DeepSeek-R1-Zero-Qwen-32B's 47 points while using 50% of the training steps [7]. This efficiency and openness is precisely what makes the elicitation question urgent: if the recipe is cheap and public, the durable value cannot be in the recipe. The recipe has since continued to commoditize — GSPO, the sequence-level objective behind the production Qwen3 models, stabilizes large-MoE RL and simplifies the infrastructure [15] — while a >400,000 GPU-hour scaling study (ScaleRL) finds that most design choices only modulate compute efficiency and that it is the recipe, not raw compute, that shifts the asymptote [16]. Compute alone does not buy the frontier; what is trained on does.

The self-improvement literature already hints at a ceiling. Self-Rewarding Language Models improve over iterations but are bounded by the model judging itself [10]. Model-collapse work shows that recursive training on generated data irreversibly erases the distribution's tails [11]. Both point to the same conclusion from different angles: durable capability gain requires an external anchor — verifiable reward or genuinely new human-grounded data — that the model did not already contain. The elicitation findings sharpen this into a quantitative test of when an RLVR signal is supplying that anchor versus merely re-weighting existing mass.

3. The Elicitation Evidence

The central datum is Shao et al.'s spurious-rewards experiment [1]. On MATH-500, RLVR lifts Qwen2.5-Math-7B by +29.1 points with ground-truth rewards, +27.1 with majority-vote pseudo-labels, +24.1 with deliberately incorrect labels, +21.4 with random rewards, and +13.8 with format-only rewards [1]. That a reward bearing no correlation — or negative correlation — with correctness recovers roughly three-quarters of the ground-truth gain is strong evidence that the policy is not learning the task from the reward; it is being nudged into a reasoning mode it already possessed. The authors trace the mechanism to 'code reasoning' — reasoning in code-like form without execution — which rises from 65% to over 90% of Qwen2.5-Math rollouts after RLVR, even under spurious rewards [1].

The model-dependence is the decisive control. The same spurious recipe yields minimal or negative changes on Llama3 and OLMo2 [1]. Only ground-truth or high-quality (majority-vote) rewards consistently help non-Qwen families. The natural reading: Qwen2.5-Math's pretraining already contains the latent reasoning circuit that any reward can surface; Llama/OLMo do not, so a meaningful signal is required. Capability that can be elicited by noise was, by definition, already there.

Yue et al. supply an independent line of evidence at the level of the capability boundary [2]. Across families and benchmarks, RLVR-trained variants win at small sampling budgets (e.g. $k=1$) but are consistently matched or overtaken by their base models as k grows; coverage and perplexity analysis shows RLVR reasoning paths already lie within the base model's sampling distribution [2]. RLVR improves the probability of reaching a solution the base could already reach — sampling efficiency — rather than enlarging the set of reachable solutions.

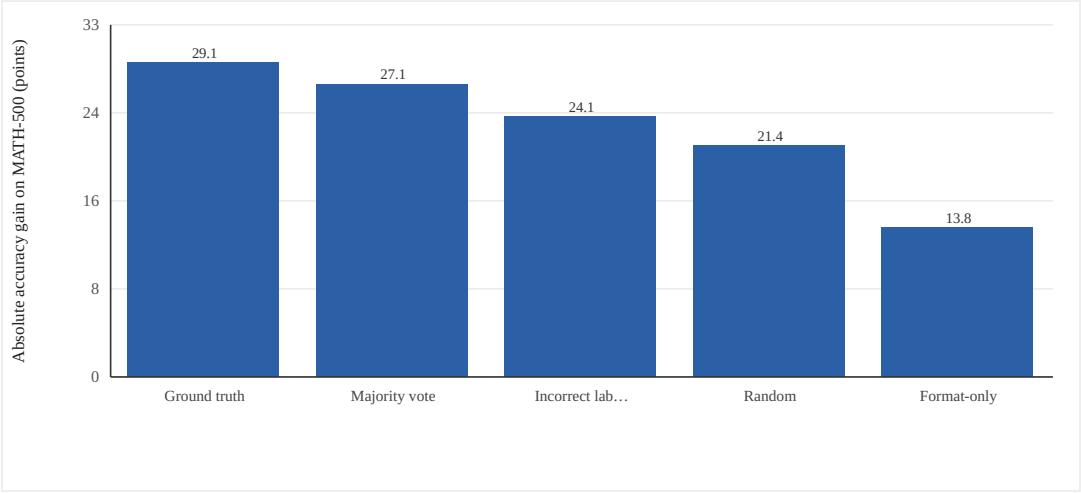


Figure 1. Spurious-reward gains on MATH-500: reward content barely matters for Qwen2.5-Math-7B. On Qwen2.5-Math-7B, random and even deliberately incorrect rewards recover ~74% and ~83% of the ground-truth gain respectively — the signature of elicitation rather than teaching. Values are absolute MATH-500 accuracy gains reported by Shao et al. [1]; all five figures verified against the primary paper. *Shao et al., Spurious Rewards: Rethinking Training Signals in RLVR, arXiv:2506.10947 (2025) [1]. · Empirical data.*

Table 1. Identical RLVR recipe, opposite outcomes by family. Qwen recovers most of the ground-truth gain from noise (elicitation); Llama/OLMo require genuine signal and gain little or are harmed by spurious rewards (no latent skill to elicit). 'n/r' = not reported as a single figure in the cell. Qwen2.5-Math-7B and -1.5B figures verified verbatim against the primary paper; Llama3/OLMo2 entries reported qualitatively in the source ('minimal or negative changes'), not as exact deltas.

Model family	Ground-truth reward	Majority-vote	Incorrect label	Random reward	Format-only
Qwen2.5-Math-7B	+29.1	+27.1	+24.1	+21.4	+13.8
Qwen2.5-Math-1.5B	n/r	n/r	+24.4	n/r	+17.6
Llama3-8B	positive (GT only)	positive	minimal / negative	minimal / negative	minimal / negative
OLMo2-7B	positive (GT only)	positive	minimal / negative	minimal / negative	minimal / negative

Shao et al., arXiv:2506.10947 [1]. 7B and 1.5B deltas verified against the paper; non-Qwen rows qualitative in source.

Table 2. The two regimes are stratified by whether target capability pre-exists in the base. Public-math/Qwen sits in elicitation; novel logic, OOD, and proprietary clinical tasks sit in teaching. Yue et al.'s crossover is reported qualitatively in the source (base matches/overtakes at large k), so no single crossover percentage is asserted.

Regime	Diagnostic signature	Representative measurement	Source
Elicitation	Base matches/overtakes RLVR at large k; spurious rewards work	Pass@k crossover across families/benchmarks (Qwen); random reward +21.4 on MATH-500	[1][2]
Teaching	Base scores ~0% at any k; only genuine reward works	Boxnet 0.00->7.91; Game-of-Life 3.49->52.29; Reasoning Gym 4.24->59.06	[3]
Metric-sensitive	Narrowing under pass@k, expansion under CoT-Pass@K	CoT-Pass@K shows correct-reasoning generalization for all K	[4]
Vertical teaching	RLVR beats SFT out-of-distribution	Med-RLVR +~8 pts over SFT on OOD evaluation (Qwen2.5-3B base)	[5]

Yue et al. arXiv:2504.13837 [2]; ProRL arXiv:2505.24864 [3]; Wen et al. arXiv:2506.14245 [4]; Med-RLVR arXiv:2502.19655 [5]. All task deltas verified.

Table 3. On a synthetic, contamination-free benchmark, the 'free' spurious-reward gains seen on public math largely vanish — only correct rewards help. This implies part of the public-benchmark elicitation effect is contamination, reinforcing the case for novel, uncontaminated vertical data. MATH-500 deltas verified [1]; clean-data column verified qualitatively against Wu et al. [12].

Reward signal	Public MATH-500 (Qwen2.5-Math-7B)	Clean RandomCalculation benchmark
Ground-truth / correct	+29.1 (large, reliable)	steady, consistent gains beyond base
Random	+21.4 (large)	unstable, no reliable improvement
Inverse / incorrect	+24.1 (large)	rapidly erodes reasoning ability

MATH-500 column: Shao et al. [1]. Clean-data column: Wu et al., Reasoning or Memorization?, arXiv:2507.10532 (2025) [12], reported qualitatively.

4. The Counter-Evidence: When RLVR Teaches

The elicitation reading is incomplete without its rebuttals, which we treat as evidence rather than noise — they delineate the conditions under which verifiable data does teach. ProRL (prolonged RL with KL control, reference-policy resets, and a diverse task suite) reports genuine boundary expansion: Nemotron-Research-Reasoning-Qwen-1.5B improves average math pass@1 by +14.7 points, code by +13.9, logic puzzles by +54.8, STEM by +25.1, and instruction-following by +18.1 over DeepSeek-R1-Distill-Qwen-1.5B [3]. Critically, on tasks where the base model scores ~0% regardless of sampling — e.g. Game-of-Life halting (3.49% → 52.29%) and the OOD Boxnet task (0.00% → 7.91%) — ProRL produces capability the base cannot reach at any k [3]. This is teaching, not elicitation.

Wen et al. address the pass@k critique directly with CoT-Pass@K, which requires both the reasoning chain and the final answer to be correct [4]. They argue plain pass@k over-credits lucky answers from flawed chains; under CoT-Pass@K, RLVR incentivizes correct reasoning for all values of K, and the effect emerges early in training and generalizes [4]. The disagreement with Yue et al. is therefore partly metric-driven: outcome-only pass@k can show narrowing while chain-correct coverage shows expansion.

The reconciliation is conditional, not contradictory. Where a competent reasoning mode is already latent (Qwen-class bases on public math), short RLVR mostly elicits and pass@k narrows; where it is absent or the task is genuinely novel (logic puzzles, OOD tasks, prolonged training), RLVR teaches and the boundary expands. The variable that decides which regime obtains is whether the capability pre-exists in the base — which is itself a function of the pretraining data.

5. Methodology: An Elicitation-vs-Teaching Diagnostic

We formalize the synthesis as a per-(base model, domain) diagnostic that estimates the teaching fraction τ of an RLVR signal — the share of its gain attributable to newly taught capability rather than elicited latent capability. τ is constructed from three measurable, source-grounded quantities, each of which is high precisely when the base lacks latent skill.

(i) Spurious-reward transfer gap, S : Run RLVR with a random/incorrect reward and with ground truth; $S = 1 - (\Delta_{\text{spurious}} / \Delta_{\text{groundtruth}})$. For Qwen2.5-Math-7B on MATH-500, $\Delta_{\text{random}}/\Delta_{\text{gt}} \approx 21.4/29.1 \approx 0.74$, so $S \approx 0.26$ — low S signals heavy elicitation [1]. For Llama3/OLMo2, spurious gains are ≤ 0 ($S \approx 1$), signalling that any gain must be taught [1]. (ii) Pass@ k crossover, C : the fraction of benchmarks where the base overtakes the RLVR model at large k . High C (Yue et al.'s results) indicates elicitation; $C \rightarrow 0$ with genuine zero-coverage solves (ProRL) indicates teaching [2][3]. (iii) Base zero-coverage rate, Z : the fraction of target tasks the base solves at $\sim 0\%$ even at large k ; Z is high in novel domains and is exactly where ProRL shows uplift [3].

We combine these as $\tau \approx \text{mean}(S, 1-C, Z)$, bounded $[0,1]$. This is an analysis construct, not a fitted model; it is deliberately transparent so that each input traces to a real measurement. The decision rule for data investment is: prioritize (base, domain) pairs with high τ , because that is where created verifiable-reward data adds capability that cannot be obtained for free by eliciting the base.

To validate the rule's direction we anchor it to the one published vertical RLVR result with an explicit teaching signature: Med-RLVR. Starting from a Qwen2.5-3B base with no reasoning supervision, RLVR on MedQA-USMLE matches SFT in-distribution but beats SFT by ~ 8 absolute points on out-of-distribution evaluation, with stepwise clinical reasoning emerging spontaneously [5]. OOD generalization that SFT cannot match is the empirical fingerprint of teaching (high τ), in a domain — clinical reasoning — under-represented in the base's pretraining.

6. Results and Cross-Family Quantification

Table 1 consolidates the spurious-reward results that drive the diagnostic. The Qwen column shows that reward content barely matters (random recovers 74% of ground-truth gain), while the cross-family rows show that on Llama3 and OLMo2 the same signals are neutral-to-harmful [1]. This swing between families under identical training is, to our knowledge, the cleanest available measurement of latent-skill dependence.

Table 2 contrasts the two regimes at the capability-boundary level. Yue et al.'s crossover (base matches or wins at large k) and ProRL's zero-coverage solves (base wins nowhere) are not contradictory once stratified by whether the task is in-distribution-latent or genuinely novel [2][3]. The ProRL logic-puzzle jump (Reasoning Gym 4.24% $\rightarrow$ 59.06%) and Game-of-Life halting (3.49% $\rightarrow$ 52.29%) are the teaching extreme; the Qwen math crossover is the elicitation extreme [2][3].

Table 3 grounds the contamination caveat. Wu et al. construct RandomCalculation, a synthetic contamination-free arithmetic benchmark, and show that on clean data only correct rewards yield reliable gains, random rewards make training unstable with no reliable improvement, and inverted rewards rapidly erode reasoning [12]. This is critical for the thesis: part of the apparent 'free' elicitation on public math may be benchmark contamination of Qwen pretraining, which only strengthens the conclusion that genuinely novel, uncontaminated vertical data is where signal is real.

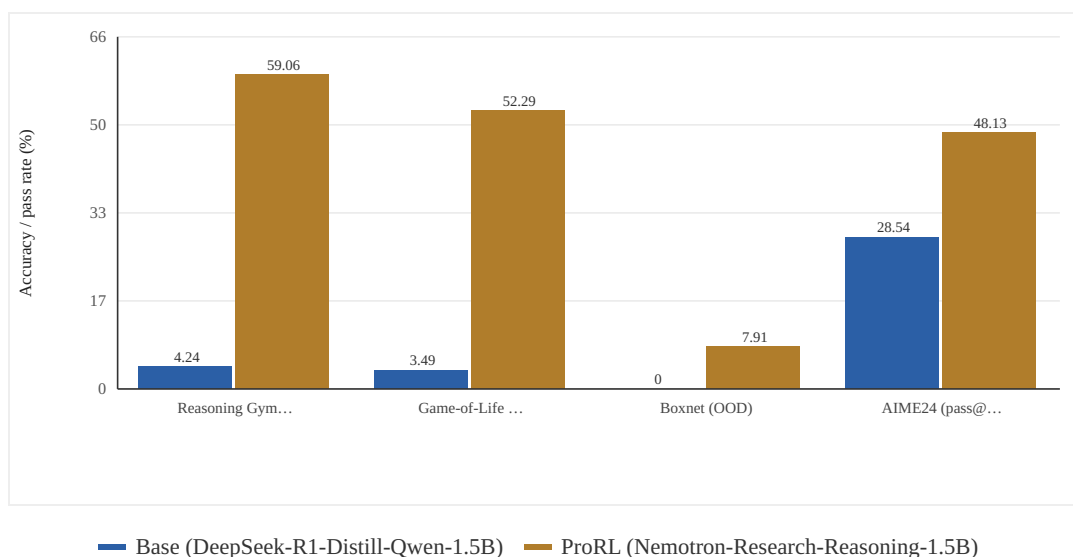


Figure 2. Teaching regime: ProRL solves tasks the base model reaches ~0% at any sampling budget. Where the base scores ~0% regardless of sampling (Boxnet 0.00, Game-of-Life 3.49), prolonged RL produces capability the base cannot reach — genuine teaching, not elicitation. All four task pairs verified against ProRL [3] and the Nemotron-Research-Reasoning-Qwen-1.5B model card. *Liu et al., ProRL, arXiv:2505.24864 (2025); figures cross-checked against the nvidia/Nemotron-Research-Reasoning-Qwen-1.5B HF model card [3]. · Empirical data.*

7. Analysis: Why the Teaching Fraction Is Highest in Verticals

The three diagnostic inputs all covary with one underlying quantity: the base model's pretraining coverage of the target domain. Public math and code are saturated in web-scale pretraining, so latent skill is high, S is low (noise suffices), C is high (base already covers the paths), and Z is near zero. The result is that an expensive verifiable environment in these domains mostly buys sampling efficiency the base could approximate for free [1][2]. This is corroborated at the environment level: RLVE shows that continuing RL on the original (saturated) task distribution yields only +0.49% absolute despite 3x more compute, whereas scaling to 400 fresh adaptive verifiable environments (RLVE-Gym) yields +3.37% on a strong 1.5B base — gains driven by environment novelty, not compute [17].

Specialized verticals invert every term. Clinical claims adjudication, regulated-industry workflows, and proprietary lab/diagnostic pipelines are sparsely represented in pretraining, so latent skill is low, ground-truth signal is required (S high), the base covers few correct chains (Z high), and there is no pass@k base to be overtaken by. Med-RLVR's OOD-over-SFT margin is the directly observed instance of this regime [5]. The marginal verifiable token is therefore worth most exactly where the open web — and thus the base model — is thinnest.

This reframes a cost question as a coverage question. The cost of building a verifiable environment is roughly domain-independent (you need a checker and tasks). The capability return is not: it scales with τ , the teaching fraction. Investing where τ is high maximizes capability-per-environment-dollar and, because the data is proprietary and uncontaminated, also maximizes defensibility — the gain cannot be replicated by a competitor eliciting a public base.

8. Implications for Data Strategy

The operational rule for an environment-building data company is to score candidate verticals by estimated τ before committing build cost. A vertical scores high when (a) the workflow throws off objective pass/fail outcomes (a checker exists), and (b) the base models demonstrably lack the capability — measurable as failure at high k, no

transfer of spurious rewards, and SFT-beating OOD gains once trained. Condition (b) is the novel filter this insight adds on top of mere verifiability.

This sharpens, and in one respect corrects, a naive 'verifiable verticals' thesis. Verifiability is necessary but not sufficient: public math and code are maximally verifiable yet low- τ because the skill is already latent and the corpus is public. The defensible targets are verticals that are simultaneously verifiable and outside the base's latent reach — proprietary clinical, insurance, and regulated workflows where the data anchor is both scarce and genuinely instructive.

For Perpetual Labs specifically, this is the evidentiary basis for steering verifiable-environment investment toward genuine knowledge frontiers rather than toward leaderboard domains. The dental claims/coding workflow is a worked high- τ candidate: a proprietary clinical record (low base coverage) paired with a deterministic CDT/X12 pass-fail outcome (a clean checker). The same logic warns against over-indexing on math/code demos, where impressive RLVR curves may reflect elicitation that competitors can reproduce for free.

9. Investment Thesis

The investable claim is narrow and falsifiable: the durable value of a verifiable-reward environment is proportional to its teaching fraction τ , and τ is highest in proprietary, under-documented verticals. This predicts that environments built on public, already-latent domains will commoditize (the recipe is open and still improving in the open [7][15][16], the base already knows it [1][2]), while environments built on high- τ proprietary verticals will retain pricing power because the capability they confer cannot be elicited from a public base.

This is a moat argument, not merely a capability argument. Two assets with identical verifiable checkers can have radically different defensibility depending on τ : a public-math environment is reproducible by anyone with the same open base; a proprietary clinical environment is not, because the teaching signal depends on data the competitor's base never saw and cannot synthesize without collapse [11]. τ thus doubles as a defensibility score.

The capital-allocation consequence is to underwrite verifiable verticals on a coverage-gap test, not a verifiability test alone. The questions for diligence are: does the base fail this domain at high k today, and is the underlying data proprietary and contamination-free? Where both answers are yes, created verifiable-reward data is a compounding, defensible asset; where either is no, it is a commodity demo.

10. Limitations and Counter-Evidence

The strongest counter is that RLVR can teach, not only elicit — ProRL's zero-coverage solves and Wen et al.'s CoT-Pass@K results are real and not explained away by elicitation [3][4]. Our position is conditional (teaching dominates where latent skill is absent), but the boundary between regimes is empirical and not yet crisply characterized; the diagnostic τ is a constructed heuristic, not a validated estimator, and its three inputs are correlated rather than independent.

Second, the elicitation evidence is concentrated on the Qwen2.5-Math family and on math benchmarks, and may be partly an artifact of benchmark contamination in Qwen pretraining rather than a general property of RLVR [12]. If so, the 'free elicitation' phenomenon is narrower than headline framing suggests — which does not weaken the strategic conclusion (target uncontaminated novel data) but does weaken any claim that RLVR is generally an elicitation trick.

Third, the single direct vertical data point (Med-RLVR) is one 3B model on medical multiple-choice [5]; its +8-point OOD margin is suggestive, not dispositive, and we lack analogous controlled measurements in claims/coding,

legal, or insurance workflows. One adjacent data point points the same way: SWE-RL, training Llama 3 on real software-evolution data with a rule-based reward, produced generalized out-of-domain reasoning gains across five held-out tasks where SFT instead degraded average performance [18] — the same RLVR-beats-SFT-out-of-distribution teaching signature, on a non-math vertical. Fourth, this analysis scores only the capability/defensibility dimension of vertical choice; it does not address willingness-to-pay, data-rights/HIPAA constraints, supply scale, or the behavior-cloning-vs-true-RL ceiling that govern whether a high- τ vertical is actually buildable. τ nominates targets; it does not clear them.

Finally, the numbers reported here are drawn from 2025 preprints; figures should be re-verified against camera-ready sources before being used in a quantitative model. Where a quantity was reported only qualitatively in the primary source (Llama/OLMo per-reward deltas, Yue et al.'s crossover magnitude, Med-RLVR absolute accuracies), we report it qualitatively rather than transcribing a fabricated cell, and flag it in the data notes.

11. Conclusion

RLVR's most striking 2025 results — gains from random rewards, pass@k narrowing — are best read not as a debunking of RLVR but as a map of where its value comes from. Where capability is latent, verifiable rewards reveal it; where capability is absent, they teach it. The teaching regime is where created data becomes a compounding, defensible asset.

That regime lives in the web's underrepresented tails: proprietary, verifiable, under-documented verticals. The practical synthesis is a single diagnostic — estimate the teaching fraction τ from spurious-reward transfer, pass@k crossover, and base zero-coverage — and invest where τ is high and the data is proprietary. This turns a live training-methods debate into a concrete data-acquisition rule, and points investment away from saturated public benchmarks toward genuine knowledge frontiers where verifiable data does what only real new data can: teach.

INVESTMENT THESIS

RLVR's headline magic — gains from random rewards on Qwen math (+21.4 pts on MATH-500, ~74% of the ground-truth +29.1; +24.1 from deliberately incorrect labels), pass@k narrowing — is largely elicitation of capability the base model already held, and it is reproducible by anyone with the same open base. The defensible, compounding asset is verifiable-reward data with a high 'teaching fraction': domains the base fails at any sampling budget (ProRL's Game-of-Life 3.49%→52.29%, Boxnet 0%→7.91%; Med-RLVR's ~+8-pt OOD win over SFT). Underwrite verifiable verticals on a coverage-gap-plus-proprietary-data test, not verifiability alone — public math/code score max verifiability but near-zero defensibility. The shortlist is proprietary, under-documented, checkable workflows (clinical claims/coding, insurance, legal adjudication), where created data teaches what no competitor's base can elicit.

References

- [1] Shao, R. et al. (2025). Spurious Rewards: Rethinking Training Signals in RLVR. arXiv:2506.10947. (Qwen2.5-Math-7B MATH-500 gains: +29.1 GT, +27.1 majority, +24.1 incorrect, +21.4 random, +13.8 format; Qwen2.5-Math-1.5B: +24.4 incorrect, +17.6 format; code-reasoning 65%→90%; spurious rewards yield minimal/negative changes on Llama3 and OLMo2.)
- [2] Yue, Y. et al. (2025). Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? arXiv:2504.13837. (Base matches/overtakes RLVR at large k across families/benchmarks; RLVR paths within base sampling distribution.)
- [3] Liu, M. et al. (2025). ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models. arXiv:2505.24864. (Nemotron-Research-Reasoning-Qwen-1.5B vs DeepSeek-R1-Distill-Qwen-1.5B: math +14.7, code +13.9, logic

- +54.8, STEM +25.1, instruction-following +18.1; Boxnet 0.00->7.91; Game-of-Life 3.49->52.29; Reasoning Gym 4.24->59.06; AIME24 28.54->48.13.)
- [4] Wen, X. et al. (2025). Reinforcement Learning with Verifiable Rewards Implicitly Incentivizes Correct Reasoning in Base LLMs. arXiv:2506.14245. (CoT-Pass@K requires both chain and answer correct; shows RLVR incentivizes correct reasoning for all K and extends the boundary on math and code.)
 - [5] Zhang, S. et al. (2025). Med-RLVR: Emerging Medical Reasoning from a 3B Base Model via Reinforcement Learning. arXiv:2502.19655. (Qwen2.5-3B base; matches SFT in-distribution on MedQA; ~+8-point accuracy gain over SFT on out-of-distribution generalization.)
 - [6] DeepSeek-AI (2025). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. arXiv:2501.12948. (R1-Zero reasoning behaviors emerge from pure RL, no SFT.)
 - [7] Yu, Q. et al. (2025). DAPO: An Open-Source LLM Reinforcement Learning System at Scale. arXiv:2503.14476. (50 points on AIME 2024 with Qwen2.5-32B base vs 47 for DeepSeek-R1-Zero-Qwen-32B, using 50% of the training steps.)
 - [8] Lightman, H. et al. (2023). Let's Verify Step by Step. arXiv:2305.20050. (Process-supervised reward model solves 78% of a representative MATH test subset, beats outcome supervision and majority voting; releases PRM800K.)
 - [9] Lambert, N. et al. (2024). Tulu 3: Pushing Frontiers in Open Language Model Post-Training. arXiv:2411.15124. (Introduces/names RLVR; scoped to checkable-answer tasks.)
 - [10] Yuan, W. et al. (2024). Self-Rewarding Language Models. arXiv:2401.10020. (Self-judging improves over iterations but is bounded by the model judging itself.)
 - [11] Shumailov, I. et al. (2023). The Curse of Recursion: Training on Generated Data Makes Models Forget. arXiv:2305.17493. (Model collapse; recursive synthetic training erases distribution tails irreversibly.)
 - [12] Wu, M. et al. (2025). Reasoning or Memorization? Unreliable Results of Reinforcement Learning Due to Data Contamination. arXiv:2507.10532. (RandomCalculation clean benchmark: only correct rewards yield steady gains beyond base; random rewards unstable with no reliable improvement; inverted rewards cause rapid collapse.)
 - [13] Singhal, K. et al. (2023). Towards Expert-Level Medical Question Answering with Large Language Models (Med-PaLM 2). arXiv:2305.09617. (Up to 86.5% on MedQA; domain corpus drives expert-level vertical capability.)
 - [14] DeepSeek-AI (2025). DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models. arXiv:2512.02556. (Scaling post-training RL compute brings V3.2 to GPT-5 parity; high-compute V3.2-Speciale surpasses GPT-5 and reaches Gemini-3.0-Pro reasoning parity, with gold-medal IMO 2025 (35/42) and IOI 2025 (492/600). <https://arxiv.org/abs/2512.02556>)
 - [15] Zheng, C. et al. (2025). Group Sequence Policy Optimization. arXiv:2507.18071. (Sequence-level importance-ratio objective; stabilizes large-MoE RL and underlies the production Qwen3 models, simplifying RL infrastructure. <https://arxiv.org/abs/2507.18071>)
 - [16] Khatri, D., Madaan, L. et al. (2025). The Art of Scaling Reinforcement Learning Compute for LLMs. arXiv:2510.13786. (>400,000 GPU-hour study fitting sigmoidal RL compute-performance curves; most design choices modulate compute efficiency while the recipe shifts the asymptote; proposes ScaleRL, validated to 100,000 GPU-hours. <https://arxiv.org/abs/2510.13786>)
 - [17] Zeng, Z., Ivison, H., Wang, Y., Yuan, L. et al. (2025). RLVE: Scaling Up Reinforcement Learning for Language Models with Adaptive Verifiable Environments. arXiv:2511.07317. (RLVE-Gym = 400 adaptive verifiable environments; joint training yields +3.37% absolute over six reasoning benchmarks from a strong 1.5B model, vs only +0.49% from continuing the original RL training despite 3x more compute. <https://arxiv.org/abs/2511.07317>)
 - [18] Wei, Y. et al. (2025). SWE-RL: Advancing LLM Reasoning via Reinforcement Learning on Open Software Evolution. arXiv:2502.18449. (Rule-based reward on software-evolution data; Llama3-SWE-RL-70B reaches 41.0% on SWE-bench Verified; RL produced generalized out-of-domain reasoning gains on five held-out tasks whereas SFT degraded average performance. <https://arxiv.org/abs/2502.18449>)